

Author Response to Reviewer 1P2E

(Rating: 4, Borderline Accept)

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We thank Reviewer 1P2E for the thoughtful and constructive feedback. Below we address each point in detail. All changes in the revised manuscript are marked in [blue text](#); line references (e.g., [L42--44]) refer to the revised `main.tex`.

R1.1: Supplement empirical findings with a theoretical discussion explaining why simple covariance invariants outperform more complex representations (bias-variance trade-off, parameterization complexity, dataset size). **Response:** We have substantially expanded the “simplicity principle” discussion in Section 6 [L615--617]. The revised text [L617] now provides an explicit parameter-count analysis: the 5 scalar invariants operate at ~ 180 samples per covariance parameter on ESOL ($N_{\text{train}} \approx 900$), whereas the lowrank variant ($k=128$) operates at ~ 7 —well below the threshold for reliable estimation. We connect this to classical model selection theory, showing that the bias-variance tradeoff penalizes higher-dimensional representations at the sample sizes typical of molecular property datasets (642–4,200 molecules). The new feature dimension ablation (Table 6, Section 5.6, [L491--515]) provides independent confirmation: even for the mean μ , RMSE monotonically degrades from $D=256$ to $D=2,048$, demonstrating that this is a fundamental statistical constraint rather than an engineering limitation.

Specific changes:

- Bias-variance theoretical analysis expanded: [L617]
- Feature dimension ablation section added: [L491--515]
- Feature dimension ablation table (Table 6): [L496--511]
- Conclusion updated with bias-variance finding: [L630]

R1.2: Conduct supplementary experiments on classification benchmarks. **Response:** We have added classification experiments on BACE (1,513 molecules, β -secretase inhibition) and BBBP (2,038 molecules, blood-brain barrier permeability) in a new Section 5.5 [L468--488]. Results (Table 7, [L473--486]) confirm the property taxonomy extends to classification: conformer features hurt BACE by -4.4% AUROC and show no benefit on BBBP ($+0.1\%$). This is consistent with the taxonomy— β -secretase binding is an electronic/binding property, and BBB permeability is governed primarily by topological features. Per-seed results are provided in Appendix K [L1056--1079].

Specific changes:

- New classification benchmark section: [L468--488]
- Classification results table (Table 7): [L473--486]
- Scope statement updated in Introduction: [L151]

- Abstract updated to include classification: [L78, 80]
- Per-seed classification appendix: [L1056--1079]

R1.3: Validate key findings on larger or industrially sourced datasets. **Response:** We acknowledge this as an important limitation in the revised Conclusion [L633]. Our benchmarks span 642 to 133K molecules across academic datasets (MoleculeNet, QM9, MARCEL); validation on larger industrially sourced datasets (ChEMBL, ZINC, proprietary ADMET panels) would strengthen the taxonomy’s generalizability. We note this as a priority for future work and have added explicit language to this effect.

Specific changes:

- Limitations paragraph updated: [L633]
- Conclusion future work: [L633]

R1.4: Increase figure resolution, standardize metrics, correct typographical errors. **Response:** We have conducted a full proofreading pass. All figures are exported at ≥ 300 DPI. Evaluation metrics are standardized (RMSE throughout for regression, AUROC for classification, [L468--488]). Typographical errors and formatting inconsistencies have been corrected across tables and main text (see blue-marked changes throughout).

Specific changes:

- Figures re-exported at ≥ 300 DPI: all `figures/*.pdf` files
- Metrics standardized: RMSE for regression, AUROC for classification [L468--488]
- Typographical corrections throughout (see blue-marked text)

R1.5: Provide clearer descriptions of scaffold split generation and number of replicates. **Response:** We have expanded Section 3.7 (Training Details, [L281]) with explicit descriptions of the Murcko scaffold split procedure (Bemis–Murcko generic scaffolds extracted via RDKit, sorted by frequency, assigned to splits maintaining 80/10/10 ratios) and seed counts (3 seeds for the main benchmark, 10 seeds for statistical validation experiments).

Specific changes:

- Scaffold split procedure details: [L281]
- Replicate counts explicitly stated: [L281]
- Stratified splits for classification explained: [L471]

Summary of changes relevant to Reviewer 1P2E:

- Added classification benchmark on BACE and BBBP [L468--488], Appendix K [L1056--1079]
- Added feature dimension ablation $D \in \{256, 512, 1024, 2048\}$ [L491--515]
- Expanded bias-variance theoretical analysis [L615--617]
- Clarified scaffold split procedure and replicate counts [L281]

- Added larger-dataset limitation acknowledgment [L633]
- All figures at ≥ 300 DPI; metrics standardized; typos corrected
- All changes marked in [blue text](#)